For instance, a commercial freezer might normally operate between -18°C and -22°C, so alerting at -15°C provides early warning of cooling system degradation before product spoilage occurs at -10°C.

- *What is the acceptable delay between a change occurring and your digital twin being updated to reflect the change?* This drives not only your choice of sensors but the entire architecture that will ingest and process the data from them. A safety-critical pressure relief system might require sub-second response times with edge processing, while tracking monthly water consumption for billing purposes tolerates daily updates transmitted over low-power networks.

3.2.2 Consider the environment

The physical environment where your sensor will operate often eliminates more options than any other factor. Start with the basic indoor/outdoor distinction: outdoor sensors need weatherproof enclosures (IP65 or higher), UV-resistant materials, and wider operating temperature ranges—my garden sensors must survive both 45°C summer days and wet winter days. Industrial environments amplify these concerns when sensors near manufacturing equipment must withstand constant vibration that would destroy consumer-grade devices within days, while dust can wreak havoc on sensitive electronics. Mounting and access is also important to consider; a sensor in a cramped ceiling space needs to be reliable enough to minimize maintenance visits, while one mounted on vibrating equipment needs secure attachment and possibly shock-dampening mounts. In figure 3.2 I use the floorplan of my home to plan the precise placement of my sensors indoors and outdoors.

NOTE

IP65 refers to an *ingress protection* code of 65, which indicates how well a device is protected against water and dust. The first digit represents solid particle protection, with **6** indicating it is protected against ingress of dust, whilst the second digit represents liquid ingress protection, with **5** indicating it can withstand powerful jets of water in any direction.